

Actual arithmetic packets through the original mixed-support construction

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1 The actual arithmetic packet through mixed-support synchronization

This calculation starts with the mixed-support construction in original shared-thread record A0961, node a7bbd422-311c-4380-ba6c-a4bdd61fce2f, chain 2032, and its full authored proof in original record A0953, node 0d2934ea-14ac-48c1-a1f3-cd918c1f5fe5, chain 1987. It does not replace the actual arithmetic function by a symmetric test polynomial. The polynomial below is the full divisor polynomial of an actual packet in the counterfactual under study.

Let Z be a nonempty finite set of actual nontrivial zeros of the original completed function $g(s) = 2\xi(s)$, with their full orders m_ρ . Retain

$$h_Z(x) = \prod_{\rho \in Z} (x - \rho)^{m_\rho}, \quad d = \sum_{\rho \in Z} m_\rho, \quad E_Z = \mathbb{C}[x]/(h_Z(x)).$$

The variable x is the original spectral variable. The additional variable t below is the cyclic coefficient coordinate; it is not substituted for x . For $a > 0$ let

$$A_a = M_{j_{h_Z}(a^s)} : E_Z \longrightarrow E_Z.$$

Here $a^s = \exp(s \log a)$ uses the real logarithm and j_{h_Z} retains all Taylor coefficients at the actual zero centres. On the summand

$$E_\rho = \mathbb{C}[z_\rho]/(z_\rho^{m_\rho}), \quad x = \rho + z_\rho,$$

this is exactly

$$A_a|_{E_\rho} = a^\rho \sum_{j=0}^{m_\rho-1} \frac{(\log a)^j}{j!} N_\rho^j, \quad N_\rho = M_{z_\rho}. \quad (1)$$

Thus no multiplicity or nilpotent term is omitted.

1.1 The original mixed-support object, with its unit

Fix $n \geq 1$. Form the source object $D_{E_Z, n} = G(E_Z)^n$ with coordinatewise addition and the source product

$$(b \star c)_k = \bigoplus_{i+j=k} b_i c_j \oplus \bigoplus_{i+j=k+n} (-1)^\bullet b_i c_j, \quad 0 \leq k < n.$$

An empty sum is τ , and the original unit is $(1^\bullet, \tau, \dots, \tau)$. Write $e = 0^\bullet_{E_Z}$. The exact amplitude target is

$$B_n = E_Z[t]/(t^n + 1), \quad \pi_n(b) = \sum_{i=0}^{n-1} p(b_i) t^i.$$

The support mask of b has its i th entry equal to one precisely when $b_i \neq \tau$. The pair consisting of the mask and $\pi_n(b)$ is injective: the monic polynomial gives the free E_Z -basis $1, t, \dots, t^{n-1}$, and the mask then distinguishes e from τ in each zero coefficient.

Let

$$E_n = (1^\bullet, e, \dots, e).$$

Multiplication by E_n sends the wholly absent input to itself and every active input to the fully supported vector with the same ordinary coefficients. To check the support assertion, fix an active input coefficient at index j . At every output index k the summand using the E_n coefficient at $k - j$ modulo n is supported; all wrap factors are supported. Thus every output coordinate is supported, including a coordinate whose resulting amplitude is zero. The amplitude assertion follows from $\pi_n(E_n) = 1$. It follows at once that $E_n^2 = E_n$ and

$$D_{E_Z, n}[E_n^{-1}] \simeq E_n D_{E_Z, n} \simeq G(B_n). \quad (2)$$

The image has unit E_n . For completeness, a unital semiring map making E_n invertible sends it to 1 by idempotence and therefore sends b and $E_n b$ to the same value. This is the localization universal property. The inclusion of the image into the original unsynchronized object is not falsely declared to preserve the original unit.

1.2 A proved split injection and its exact retraction

Define

$$\iota_n : E_Z \longrightarrow B_n, \quad f \longmapsto f, \quad r_n : B_n \longrightarrow E_Z, \quad b \longmapsto \frac{1}{n} \text{Tr}_{B_n/E_Z}(M_b).$$

The first is a unital ring homomorphism. The second is an E_Z -linear map of modules; it is not asserted to be multiplicative. In the basis $1, t, \dots, t^{n-1}$, multiplication by t^k has no diagonal entries for $1 \leq k < n$, since it shifts every basis index by the nonzero class $k \pmod{n}$. Its wrap entries carry the original minus signs. Multiplication by $b_0 \in E_Z$ is the scalar matrix with n diagonal entries b_0 . Consequently

$$\text{Tr}_{B_n/E_Z} \left(M_{\sum_{i=0}^{n-1} b_i t^i} \right) = n b_0, \quad r_n \left(\sum_{i=0}^{n-1} b_i t^i \right) = b_0, \quad r_n \iota_n = 1_{E_Z}. \quad (3)$$

In particular the trace factor n and its inverse in the retraction are both retained. No arithmetic trace form has been rescaled.

For a linear map $T : M \rightarrow N$ over E_Z , its split lift is $T^\tau(m^\bullet) = T(m)^\bullet$ and $T^\tau(\tau) = \tau$. It is $G(E_Z)$ -linear: addition of supported inputs is the original module addition, scalar multiplication is the original module action, and the cases containing an absent scalar or input are absorbed by τ . A zero result is $(0_N)^\bullet$, not τ . Thus (3) lifts to an actual retract

$$E_Z^\tau \xrightarrow{\iota_n^\tau} G(B_n) \xrightarrow{r_n^\tau} E_Z^\tau, \quad r_n^\tau \iota_n^\tau = 1. \quad (4)$$

Through (2), the first map is the fully supported mixed vector $(f^\bullet, e, \dots, e)$ for supported f , and the wholly absent vector for τ . Therefore every nonzero amplitude packet class remains nonzero through this specific mixed-support realization and localization.

Write $\sigma_Z : E_Z \hookrightarrow Q$ and $J_Z : Q \rightarrow E_Z$ for the original arithmetic packet injection and observation, whose proved identity is $J_Z \sigma_Z = 1_{E_Z}$. Their composite with the maps just computed is

$$\sigma_Z r_n : B_n \longrightarrow Q, \quad (\sigma_Z r_n) \iota_n = \sigma_Z.$$

This is a composite of the established maps, not a new assertion that the spectral coefficients are unrelated to arithmetic cohomology.

The scaling intertwiner used here comes from the original source. Retain $Q = \mathcal{B}/\Theta V$, $D = -x\partial_x$, and $\Theta\phi(x) = \sum_{j \in \mathbb{Z} \setminus \{0\}} \phi(jx)$. On both original test spaces define

$$(\mathcal{R}_a F)(x) = F(x/a), \quad a > 0.$$

Positive dilation preserves even Schwartz functions and their two zero moments: the value at zero is unchanged and the integral is multiplied by a . Direct substitution in the theta sum proves $\Theta\mathcal{R}_a = \mathcal{R}_a\Theta$. Thus $\mathcal{R}_a$ induces the original action $\overline{\mathcal{R}}_a$ on Q . Substitution $x = ay$ in the Mellin integral gives

$$\mathcal{M}(\mathcal{R}_a F)(s) = a^s \mathcal{M}F(s), \quad J_Z \overline{\mathcal{R}}_a = A_a J_Z.$$

The source's complete finite-packet theorem identifies $\sigma_Z(E_Z) = Q[h_Z(D)]$ and makes J_Z its inverse there. Since $\mathcal{R}_a D = D\mathcal{R}_a$, that subspace is invariant. Applying the bijection J_Z to its two vectors therefore proves

$$\overline{\mathcal{R}}_a \sigma_Z = \sigma_Z A_a.$$

This carries the full factor a^ρ and every Taylor coefficient in (1) from the actual theta quotient.

1.3 The exact coefficient Frobenius comparison

Let q be an odd prime power with $\gcd(q, n) = 1$. On the integral coefficient algebra

$$C_{n, \mathbb{Z}} = \mathbb{Z}[t]/(t^n + 1)$$

define the unital algebra map

$$\Psi_q(t) = t^q, \quad \Psi_q(c) = c \quad (c \in \mathbb{Z}).$$

It is well-defined since $(t^q)^n = (-1)^q = -1$. Because $\gcd(q, 2n) = 1$, choose $r > 0$ with $q^r \equiv 1 \pmod{2n}$. The identity $t^{2n} = 1$ then gives $\Psi_q^r = 1$, proving it is an automorphism. Its matrix in the original coefficient basis is the signed permutation

$$t^i \longmapsto (-1)^{\lfloor qi/n \rfloor} t^{qi \bmod n}. \quad (5)$$

Reduction of this integral map to $\mathbb{F}_q$ is exactly the source's q -power Frobenius on $\mathbb{F}_q[t]/(t^n + 1)$, because every scalar of $\mathbb{F}_q$ has q th power equal to itself. Its extension to $\mathbb{C}$ is a coefficient-linear lift of that integral matrix. It is not the map $c \mapsto c^q$ on complex scalars. These two specializations are linked by the displayed integral map; no ring homomorphism $\mathbb{F}_q \rightarrow \mathbb{C}$ is asserted.

There is also a map on the full unsynchronized mixed-support object: send coefficient b_i to $((-1)^\bullet)^{\lfloor qi/n \rfloor} b_i$ in slot $qi \bmod n$, fixing E_Z scalars and preserving absent inputs. It is additive. On two supported monomials, reducing the exponent before or after the map gives the same wrap factor because $((-1)^\bullet)^q = (-1)^\bullet$; the absent cases are absorbed, and distributivity proves the product identity for arbitrary vectors. Hence this is a semiring morphism, fixes $\mathbf{E}_n$, preserves every mask up to the stated permutation, and induces Ψ_q through (2).

Extend Ψ_q E_Z -linearly to B_n . It fixes the subspace $\iota_n(E_Z)$ pointwise. It preserves r_n : formula (5) moves no index $i \in \{1, \dots, n-1\}$ to zero modulo n , and fixes index zero. Hence

$$\Psi_q \iota_n = \iota_n, \quad r_n \Psi_q = r_n. \quad (6)$$

Every eigenvalue of the complex coefficient operator Ψ_q is a root of unity, because $\Psi_q^r = 1$. Over $\mathbb{C}$ the polynomial $X^r - 1$ has no repeated roots, so the operator is diagonalizable. This is the exact finite coefficient purity calculation transported through its integral matrix.

1.4 The original arithmetic action, full jets, and modulus

Extend A_a to B_n by coefficient multiplication:

$$A_a^B \left(\sum b_i t^i \right) = \sum A_a(b_i) t^i.$$

Since A_a is multiplication by $j_{h_Z}(a^s) \in E_Z$, this commutes with the E_Z -linear automorphism Ψ_q . The maps in (3) intertwine the arithmetic action exactly:

$$A_a^B \iota_n = \iota_n A_a, \quad r_n A_a^B = A_a r_n.$$

Consequently they also intertwine A_a with $A_a^B \Psi_q$ because of (6). On the coefficient eigenspace of Ψ_q with eigenvalue ω , the original ρ -packet action is

$$\omega a^\rho \sum_{j=0}^{m_\rho-1} \frac{(\log a)^j}{j!} N_\rho^j. \quad (7)$$

The entire nilpotent polynomial is retained. For $a \neq 1$, its nilpotent part has the same nilpotence index as N_ρ :

$$\exp((\log a) N_\rho) - 1 = N_\rho \left((\log a) I + \sum_{j=2}^{m_\rho-1} \frac{(\log a)^j}{j!} N_\rho^{j-1} \right),$$

and the factor in parentheses is invertible because its constant coefficient is nonzero. The expression also states the case $m_\rho = 1$ directly.

Let e_ρ be the full Chinese-remainder idempotent in E_Z . The vector

$$v_\rho = e_\rho(x - \rho)^{m_\rho-1}$$

is nonzero: in its ρ -local quotient its coefficient of $z_\rho^{m_\rho-1}$ is one. It is killed by N_ρ and satisfies $A_a v_\rho = a^\rho v_\rho$. The vector $\iota_n(v_\rho)$ is nonzero by the retraction, lies in the fixed coefficient line of Ψ_q , and is therefore an eigenvector of $A_a^B \Psi_q$ with the unchanged eigenvalue a^ρ .

Suppose now the actual counterfactual supplies $\rho = \frac{1}{2} + \delta + i\gamma$, $0 < \delta < \frac{1}{2}$. For every $a > 1$,

$$\frac{|a^\rho|^2}{a} = a^{2\delta} > 1, \quad \log \frac{|a^\rho|^2}{a} = 2\delta \log a. \quad (8)$$

For the other coefficient eigenspaces in (7), $|\omega| = 1$ gives exactly the same modulus. At $a = q$, these eigenvalues are $q^\rho \omega$. The coefficient pure spectrum therefore preserves the actual arithmetic weight defect in this specified construction.

Take every tensor power in this paragraph over $\mathbb{C}$. The vector $\iota_n(v_\rho)^{\otimes_{\mathbb{C}} k}$ is nonzero because $(r_n)^{\otimes_{\mathbb{C}} k}$ sends it to $v_\rho^{\otimes_{\mathbb{C}} k}$. The latter is nonzero: a complex linear functional taking v_ρ to 1 has k -fold complex tensor power taking that tensor to 1. Its eigenvalue under $(A_a^B \Psi_q)^{\otimes_{\mathbb{C}} k}$ is $a^{k\rho}$, and

$$\log \frac{|a^{k\rho}|^2}{a^k} = 2k\delta \log a.$$

This is an exact input to the user's selected Deligne amplification argument. The computation does not import a cohomological upper estimate from another object. It identifies which actual nonzero class and which unchanged eigenvalue such an estimate must control.

1.5 The trace obstruction also survives the pure coefficient factor

Assume Z is stable under $\rho \mapsto 1 - \bar{\rho}$ and under conjugation, with the original multiplicities. Retain the actual derivative-residue trace contraction

$$\mathcal{T}_Z(f, u) = \sum_{\rho \in Z} m_\rho \overline{f(1 - \bar{\rho})} u(\rho).$$

This formula is the contraction of the original residue pairing using g' ; the factors in $g = 2\xi$ are not modified here. In a genuine off-line quartet let m be its common multiplicity and set $f = e_\rho - e_{1-\bar{\rho}}$. The two nonzero evaluations contribute $-m$ each, so $\mathcal{T}_Z(f, f) = -2m$.

On $C_n = \mathbb{C}[t]/(t^n + 1)$ retain the positive form

$$H_n(c, d) = \sum_{j=0}^{n-1} \overline{c(r_j)} d(r_j), \quad r_j = \exp\left(\frac{(2j+1)\pi i}{n}\right).$$

The distinct roots make evaluation an isomorphism $C_n \simeq \mathbb{C}^n$, proving positive definiteness. The operator Ψ_q permutes the evaluations, so it preserves H_n . The original constant vector has $H_n(1, 1) = n$. Under $B_n \simeq E_Z \otimes_{\mathbb{C}} C_n$ the tensor form therefore satisfies

$$(\mathcal{T}_Z \otimes_{\mathbb{C}} H_n)(\iota_n f, \iota_n f) = -2mn < 0.$$

This proves the actual negative trace vector survives this exact positive coefficient attachment, with the factor n retained. It does not assert that the global arithmetic polarization is automatically this tensor form. Any different global comparison has its own map and must be calculated on this surviving input.